

3D-printed cube dials into energy harvesting

As Internet-connected smart devices become smaller and more widespread, the challenge of powering these grows more acute. An inexpensive energy harvester developed at King Abdullah University of Science and Technology (KAUST) can help recharge Internet of Things (IoT) gadgets using radio waves from wireless sources.

One of the ways researchers are miniaturising devices for IoT applications is through an approach called system-on-package. Recent work has shown that the protective packaging around microelectronic devices could be used to accommodate components, such as communication antennae, for significantly reduced costs and space requirements.



Symmetric geometry of the cube makes it more effective at gathering energy

Atif Shamim, professor of electrical engineering and an expert in energy harvesting, realised that system-on-package principles could help IoT devices become more self-sufficient. His team investigated strategies to build highly compact antennae that can tune into radio frequency (RF) signals emitted from mobile and wireless devices. They then teamed up with Khaled Salama's group at KAUST to convert this energy into electricity using semiconductor diodes.

Most RF harvesters can only tap into a single part of the wireless spectrum, such as 3G standard. Shamim's team, however, aimed to produce a multi-band device that could accumulate more energy from multiple sources of communication.

The researchers turned to a cube-shaped package and the mathematical concept of fractals—patterns that repeat from small to large scale—to build the harvester. First, the team 3D-printed a square plastic substrate and then screen-printed fractal antennae on its surface using silver metal. Finally, it glued five of the plastic pieces together to form a cube, roughly five centimetres in size.

Fractal antennae can introduce multiple resonances that allow access to broader parts of the radio spectrum. Symmetric geometry of the cube worked to enhance this effect by gathering radiation all around the cube. Subsequent wireless spectrum scanning revealed several distinct frequencies where energy harvesting could work.

Flags that can generate energy from wind and sun

Scientists from University of Manchester have created flags that can generate electrical energy using wind and solar power. The novel wind and solar energy-harvesting flags have been developed using flexible piezoelectric strips and flexible photovoltaic cells. Piezoelectric strips allow the flags to generate power through movement, whilst photovoltaics is the best-known method of harnessing electric power using solar cells. This study is the first to simultaneously harvest wind and solar energies using inverted flags.

The energy-harvesting flags are capable of powering remote sensors and small-scale portable electronics that can be used for environmental sensing, such as to monitor pollution, sound levels and heat.

According to Jorge Silva-Leon, who is from Manchester's School of Mechanical, Aerospace and Civil Engineering and the lead author of the study, under the action of wind, the flags bend from side to side in a repetitive fashion—also known as limit-cycle oscillations. This makes them perfectly suited for uniform power generation from the deformation of piezoelectric materials. Simultaneously, the solar panels bring a double benefit by acting as a destabilising mass that triggers the onset of flapping motions at lower wind speeds and, of course, generating electricity from ambient light.

Dr Andrea Cioncolini, co-author of the study, adds, "Wind and solar energies typically have intermittencies that tend to compensate each other. The Sun does not usually shine during stormy conditions, whereas calm days with little wind are usually associated with shiny Sun. This makes wind and solar energies particularly well-suited for simultaneous harvesting, with a view at compensating their intermittency."

The aim of the study is to allow cheap and sustainable energy-harvesting solutions that can be deployed and left to generate energy with little or no need for maintenance.

AI Pictionary game that lets humans and AI play together

Allen Institute for Artificial Intelligence has released Iconary, an AI Pictionary game that allows a human to collaborate with artificial intelligence (AI) software. Iconary showcases the world's first AI system capable of playing a Pictionary-style drawing and guessing game together with a human partner. In Iconary, the human teams up with AllenAI to either draw or guess a set of unique phrases. There is a limited set of icons available to compose the drawing.

AllenAI combines advanced computer vision, language understanding and common sense reasoning to make guesses based on the drawings and to produce its own complex scenes for the human to try to guess. Iconary is the one of the first times an AI system is paired in a collaborative game with a human player instead of antagonistically working against them. AllenAI is capable of both understanding and producing a nearly infinite combination of real-world scenarios represented in the phrases in Iconary—a compelling example of the potential of common sense for AI and the power of human-AI collaboration.